

1) Unit 4: Rotational Kinematics

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1) $B: 45 \text{ rpm}, 4.7 \text{ radians/seconds}$

2) $D: 2.2 \text{ m/s}^2$

3) $D: 10 - 10 \text{ W}$

4) $B: 0.53 \text{ N}$

5) $C: 42 \text{ m/s}$

2) Unit 5: Work and Energy

1) $D: 200 \text{ Nm}, 37 \text{ degrees}$

2) $\text{Work done} = F \times s$
 $= 1000 \text{ N} \times 0.1 \text{ m} = 100 \text{ J}$

b) change in kinetic energy = work done (according to the work-energy theorem)
 $= 100 \text{ J}$

c) $KE = \frac{1}{2} m v^2$
 $= 100 = \frac{1}{2} (0.15) v^2$
 $v^2 = \frac{200}{0.15} = 1333.33$

$v = \sqrt{1333.33} = 36.5 \text{ m/s}$

$$d) R = \frac{v^2 \sin(2\theta)}{g} \quad \sin(90^\circ) = 1$$

$$R = \frac{v^2}{g} = \frac{(36.5)^2}{9.8} = \frac{1332.25}{9.8} = 136 \text{ m}$$

$$3) a) F_k = \mu_k \cdot F_N$$

$$F_k = \mu_k \cdot m \cdot g$$

$$= 0.05 \times 40 \text{ kg} \times 9.8 \text{ m/s}^2$$

$$F_k = 0.05 \cdot 40 \cdot 9.8 = 19.6 \text{ N}$$

$$W = F_k \cdot d$$

$$W = 19.6 \cdot 40 = 784 \text{ J}$$

$$b) P = \frac{W}{t}$$

$$= \frac{784}{30} = 26.13 \text{ W}$$

$$4) D: \$233$$

$$3) \text{ Unit 6: } \underline{L} \text{ linear momentum}$$

$$2) C: 0 \text{ m/s}^{-1}$$

$$2) P = mv$$

$$= (7 \times 10^{-25}) \cdot 350 = 7 \times 10^{-23} \text{ kg} \cdot \text{m/s}$$

$$P_{\text{total}} = \sqrt{P^2 + P^2} = \sqrt{2P^2} = P\sqrt{2}$$

$$P_{\text{total}} = (7 \times 10^{-23}) \cdot \sqrt{2} = 9.899 \times 10^{-23} \text{ kg} \cdot \text{m/s}$$

$$V_f = \frac{P_{\text{total}}}{m_{\text{total}}} = \frac{9.899 \times 10^{-23}}{40 \times 10^{-25}} = 247.5 \text{ m/s}$$

$$3) a) J = F \cdot t$$

$$= 400 \cdot 0.200 = 800 \text{ N}\cdot\text{s}$$

$$b) J = \Delta p = m \cdot (v_f - v_i)$$

$$v_f = v_i + \frac{J}{m}$$

$$v_f = 2.80 + \frac{800}{200} = 2.80 + 4.00 = 6.80 \text{ m/s}$$

4) c: elastic

$$5) a) m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$v_f = \frac{(30,000)(0.850) + (110,000)(0)}{30,000 + 110,000}$$

$$v_f = \frac{25,500}{140,000} = 0.182 \text{ m/s}$$

$$b) \Delta KE = KE_{\text{initial}} - KE_{\text{final}}$$

$$KE_{\text{initial}} = \frac{1}{2} m_1 v_1^2 = \frac{1}{2} (30,000)(0.850)^2 = 10,837.5$$

$$KE_{\text{final}} = \frac{1}{2} (m_1 + m_2) v_f^2 = \frac{1}{2} (140,000)(0.182)^2 = 2,319.88$$

$$\Delta KE = 10,837.5 - 2,319.88 = 8,517.62 \text{ J}$$

4) Unit 7: Statics

1) A: remain motionless

$$2) W = m \cdot g = 20.0 \cdot 9.8 = 196 \text{ N}$$

$$T_{\text{plank}} = -(100 \cdot 196 \text{ J}) = -19600 \text{ N}\cdot\text{m} = 196 \text{ N}$$

$$T_{f_2} = 0.50 \cdot f_2$$

$$0.50 \cdot f_2 - 196 = 0$$

$$f_2 = \frac{196}{0.50} = 392 \text{ N}$$

$$f_2 = f_2 - 196 = 392 - 196$$

$$f_2 = 196 \text{ N}$$

$$f_2 = 392 \text{ N}$$

5) Unit 8: Rotational Dynamics

$$1) a) I = \frac{1}{2}MR^2$$

$$I = \frac{1}{2}(2)(0.08)^2$$

$$= \frac{1}{2}(0.0064) = 0.0032 \text{ kgm}^2$$

$$b) \tau = I\alpha$$

$$\tau = (0.0032)(2) = 0.0064 \text{ Nm}$$

$$\tau = 0.0064 \text{ Nm}$$

$$2) a) \vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.2 & 0.2 & 0 \\ 12 & 0 & 0 \end{vmatrix}$$

$$\vec{\tau} = \hat{i} \begin{vmatrix} 0.2 & 0 \\ 12 & 0 \end{vmatrix} - \hat{j} \begin{vmatrix} 0.2 & 0 \\ 12 & 0 \end{vmatrix} + \hat{k} \begin{vmatrix} 0.2 & 0.2 \\ 12 & 0 \end{vmatrix}$$

$$\vec{\tau} = 0\hat{i} - 0\hat{j} + \hat{k} \cdot ((0.2)(0) - (0.2)(12))$$

$$\vec{\tau} = -2.4\hat{k} \text{ Nm}$$

$$b) \vec{\tau} = \vec{r} \times \vec{F} = +2.4\hat{k} \text{ Nm}$$